

Jaemin Kim

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RESEARCH INTEREST

Aiming to improve the controllability of advanced generative models to **maximize rewards**. Recently contributed to this goal by exploring efficient guidance methods, such as training-free [C2,C4,C5], Plug-and-Play [C3, C6], and RL-based agentic search [W1, P2].

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) <i>Ph.D. in Artificial Intelligence</i>	Mar 2025 - Now <i>Advisor: Prof. Jong Chul Ye</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>M.S. in Artificial Intelligence</i>	Mar 2023 - Feb 2025 <i>Advisor: Prof. Jong Chul Ye</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>B.S. in Bio and Brain Engineering, Electrical Engineering (Double Major)</i>	Mar 2016 - Feb 2023 <i>GPA: 3.96/4.3 (Magna Cum Laude)</i>

PUBLICATIONS *: Equal Contribution

[C6] <u>Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs</u> Jaemin Kim*, Hangeol Chang*, Hyunmin Hwang*, Choonghan Kim, Jong Chul Ye	<i>ICML 2026</i>
[C5] <u>Training-Free Reward-Guided Image Editing via Trajectory Optimal Control</u> Jinho Chang*, Jaemin Kim*, Jong Chul Ye	<i>ICLR 2026</i>
[C4] <u>Free²Guide: Training-Free Text-to-Video Alignment using Image LVLM</u> Jaemin Kim, Bryan Sangwoo Kim, Jong Chul Ye	<i>ICCV 2025</i>
[C3] <u>Optical-Flow Guided Prompt Optimization for Coherent Video Generation</u> Hyelin Nam*, Jaemin Kim*, Dohun Lee, Jong Chul Ye	<i>CVPR 2025</i>
[C2] <u>Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI</u> Won Jun Kim*, Hyungjin Chung*, Jaemin Kim*, Sangmin Lee, Byeongsu Sim, Jong Chul Ye	<i>CVPR 2025</i>
[C1] <u>Generalized Consistency Trajectory Models for Image Manipulation</u> Beomsu Kim*, Jaemin Kim*, Jeongsol Kim, Jong Chul Ye	<i>ICLR 2025</i>

PRE-PRINTS AND WORKSHOPS *: Equal Contribution

[W1] <u>AgentPSO: Evolving Agent Reasoning Skill via Multi-agent Particle Swarm Optimization</u> Hyunmin Hwang*, Jaemin Kim*, Choonghan Kim, Hangeol Chang, Jong Chul Ye	<i>ICML AI4Math 2026</i>
[P4] <u>Adaptive Guidance for Retrieval-Augmented Masked Diffusion Models</u> Jaemin Kim and Jong Chul Ye	<i>Preprint 2026</i>
[P3] <u>DMD-augmented Unpaired Neural Schrödinger Bridge for Ultra-Low Field MRI Enhancement</u> Youngmin Kim*, Jaeyun Shin*, Jeongchan Kim*, Taehoon Lee, Jaemin Kim, Peter Hsu, Jelle Veraart, Jong Chul Ye	<i>Preprint 2026</i>

- [P2] **Dementia-R1: Reinforced Pretraining and Reasoning from Unstructured Clinical Notes for Real-World Dementia Prognosis** *Preprint 2026*
 Choonghan Kim*, Hyunmin Hwang*, Hangeol Chang*, **Jaemin Kim***, Jinse Park,
 Jae-Sung Lim, Jong Chul Ye
- [P1] **HiCBridge: Resolution Enhancement of Hi-C Data Using Direct Diffusion Bridge** *Preprint 2024*
Jaemin Kim* and Jong Chul Ye

EXPERIENCES

Research Intern <i>Sony Group - Creative AI Lab</i>	May 2026 - Oct 2026 <i>Tokyo Osaka, Japan</i>
Karlsruhe Institute of Technology (KIT) <i>Electrical Engineering and Information Technology (Exchange Student)</i>	Sep 2019 - Feb 2020 <i>Karlsruhe, Germany</i>

PATENTS

Method and Apparatus for Approximating Gradient of Artificial Neural Network Model • (South Korea) Patent No.10-2025-0145337	2025
A Single, Composable Plug-and-play Reasoner for Frozen LLMs • (South Korea) Patent No.10-2025-0132734	2025

HONORS AND AWARDS

32nd Samsung Humantech Paper Award <i>Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs</i>	Silver Prize <i>Track: Signal Processing</i>
Dongwon-KAIST scholarship <i>Full scholarship for tuition and stipend</i>	Mar 2023 - Feb 2025
Baden-Württemberg Stipendium <i>Scholarship for international students to study at a university in Baden-Württemberg</i>	Sep 2019 - Feb 2020
South Korea National Science & Technology Scholarship <i>Scholarship for attracting outstanding talents</i>	Mar 2018 - Mar 2020

ACADEMIC ACTIVITIES

Reviewer <i>ICLR 2026, ICML 2026, NeurIPS 2026, TMLR</i>	
Persident, Student Council <i>KAIST AI</i>	May 2025 - Mar 2026
Teaching Assistant <i>KAIST - AI618: Generative Models and Unsupervised Learning</i>	Mar 2025 - Jun 2025

REFERENCES

Cristian Simon <i>u6562565@anu.edu.au</i>	Mentor at Sony <i>AI Researcher, Creative AI, Sony Group</i>
Jong Chul Ye <i>jong.ye@kaist.ac.kr</i>	M.S., Ph.D. Advisor <i>Endowed Chair Professor, Kim Jaechul Graduate School of Artificial Intelligence, KAIST</i>